

Natural Language Processing

CSE5NLP

SUBJECT LEARNING GUIDE

Teaching Period Term 3, 2024

Subject Coordinator

Dr Chang Liu

SUBJECT DETAILS

Subject Code:	CSE5NLP
Subject Title:	Natural Language Processing
Teaching Period:	Term 3, 2024
Location(s):	Bundoora
Credit Points:	15
Delivery Mode:	Online
AQF Level:	8

ENROLMENT REQUIREMENTS

Prerequisites:	CSE5AIF
Co-requisites:	None
Assumed Skills & Knowledge:	Students are expected to have done subjects on programming with Python machine learning. They must have familiarity with basic probability and statistics.
Special Study Requirements:	Students with special needs whose participation in this subject might be limited by any study requirements are encouraged to consult the Equality and Diversity Centre (EDC). To contact the Centre: • Telephone on (03) 9479 2900 (Melbourne); (03) 5444 7410 (Bendigo); (02) 6024 9628 (Albury-Wodonga) • National Relay Service Deaf and Hearing Impaired: 133 677 (within Australia only) Email: equality@latrobe.edu.au or refer to the website.
Additional subject requirements:	Internet access. All resources are available online via the LMS.
Class materials:	All materials are available online via the LMS.

STAFF CONTACTS

Subject Coordinator:	Dr Chang Liu
Email:	C.Liu6@latrobe.edu.au
Tel:	On request
Location:	
Consultation time/s:	Times by email and/or appointment

ABOUT THIS SUBJECT

Natural language processing (NLP) is broadly concerned with the interactions between computers and natural (i.e. human) languages; more particularly, it is concerned with the question of how to program

computers to process and analyse large amounts of natural language data. Following a review of the essential mathematical and linguistic concepts underlying natural language processing, you will develop skills in important natural language processing sub-tasks including accessing corpora, tokenisation, morphological analysis, word sense disambiguation, part-of-speech tagging and sentence structure analysis. You will then apply these skills in the context of applications such as text categorisation, text clustering, text recommendation and information retrieval. Where appropriate, both lexical (i.e. dictionary-based) and machine learning approaches will be used.

SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

1. Apply natural language processing sub-tasks — including tokenisation, morphological analysis, word sense disambiguation, part-of-speech tagging and sentence structure analysis — to natural languages texts
2. Describe and evaluate the methods and algorithms used to process different types of textual data
3. Devise natural language processing (NLP) pipelines using existing NLP code libraries, text corpora and lexical resources such as WordNet
4. Critically evaluate results of applying natural language processing methods to real-world tasks such as text categorisation, text clustering, text recommendation and information retrieval.

GRADUATE CAPABILITIES (GC)

GC	GC Point
Communication	Midpoint
Inquiry and Analysis	Midpoint
Personal and Professional	Midpoint
Discipline Knowledge and Skills	Midpoint

LEARNING ACTIVITIES

Timetable details are available at <https://www.latrobe.edu.au/students/your-course/timetables>. Students should also regularly check this subject's LMS site for updated information.

Session	Date	Learning Activities	Contact hours
1	Monday 22 April, 2024	Introduction to NLP and NLP basic tasks	3
2	Monday 29 April, 2024	Text classification	3
3	Monday 06 May, 2024	Sequence labelling, vector semantics and embeddings	3
4	Monday 13 May, 2024	Neural networks and neural language models	3
5	Monday 20 May, 2024	Deep learning architectures for sequence processing and transfer learning	3

6	Monday 27 May, 2024	NLP on Azure	3
7	Monday 03 June, 2024	Exam week	
		Total	18

Details of any self-allocation activities
N/A

SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately **15 hours per week of self-directed study in this subject**. This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

LEARNING RESOURCES & TECHNOLOGIES

Resource Type	Title	Author and Year	Publisher
Required			
All materials are available online via the LMS.			
Recommended			
All materials are available online via the LMS.			

ASSESSMENT AND FEEDBACK SUMMARY

Week	Date due	%	Assessment	Key Criteria	Feedback method	SILOs assessed
3	Sunday, 12 May 2024 by 23.59 (Melbourne time)	20	Assessment 1: Implementing solutions using Python — part 1	- Build a chatbot that is smart and intuitive by applying natural language processing sub-tasks (30%) - Apply author attribution using the n-gram language model to identify authors (40%)	Written feedback will be provided by Sunday at the end of Week 4, 23:59	1, 2, 3, 4

				• Apply natural language processing algorithms accurately to classify fake news versus real news (30%)		
5	Sunday, 26 May 2024 by 23.59 (Melbourne time)	30	Assessment 2: Implementing solutions using Python — part 2	• Apply natural language processing sub-tasks such as TF-IDF to clean the dataset provided (30%) • Implement word-embedding for the purposes of classification to derive accuracy and F1 score of the dataset provided (20%) • Apply natural language processing sub-tasks to enable robots and chatbots to return specific commands accurately (30%) • Critically evaluate the results of the chatbot program and report limitations (20%)	Written feedback will be provided by Sunday at the end of Week 6, 23:59	1, 2, 3, 4
7	TBA (centrally scheduled)	50	Assessment 3: Final exam	• Apply natural language processing sub-tasks to real-world problems, and critically evaluate the methods, algorithms and the results of	Automated feedback will be provided once quiz is submitted in the LMS	1, 2, 3, 4

				applying natural language processing by answering questions accurately (100%)		
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Additional information about assessment & feedback

Students should keep a copy of all assignments submitted for assessment.

MANDATORY REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Mandatory Requirements** may be built into the assessment of the learning outcomes of your subject.

There are no Hurdle Requirements for this subject.

GENERAL ASSESSMENT REQUIREMENTS

SUBMITTING WRITTEN WORK WITH TURNITIN

Turnitin is a web-based text-matching software system used at La Trobe University to assist you in writing your assignments and checking for similarity with existing published work. Please remember to allow adequate time to submit your assignment to Turnitin. **It is your responsibility to have your assignment submitted by the assessment due date.** Not receiving an originality report is not an acceptable reason for requesting an assignment extension unless it is a required part of the assessment.

POLICIES, PROCEDURES AND GUIDELINES

The University has a comprehensive policy framework to which both staff and students must adhere. You should familiarise yourself with those policies, procedures and guidelines likely to affect you, especially the following:

- [Academic Progress Review Policy](#)
- [Assessment Policy](#)
- [Health and Safety](#) (OHS)
- [Privacy](#)
- [Student Academic Misconduct Policy](#)

ADJUSTMENTS TO ASSESSMENT

You may be affected by a range of adverse circumstances while you are preparing for or undertaking an assessment task. There are avenues for adjustments to your assessment depending on the circumstances you face including opportunities to apply for extensions to submission dates, Special Consideration and Alternative Assessment Schedules. For more detailed information refer to the [Assessment Procedure – Adjustments](#).

REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT TASK

Where you know in advance that you will not be able to submit an assessment task by the due date due to adverse circumstances that have affected you during the preparation of the task, you will need to [request an extension of time to submit](#). **This must be done at least three days prior to the due date.** Penalties normally apply if you submit an assessment after the due date. **Poor time management is not an acceptable reason for an extension.**

To apply, go to <https://www.latrobe.edu.au/students/admin/forms/request-an-extension/request>

SPECIAL CONSIDERATION

If you have experienced serious short term, adverse and unforeseen circumstances that substantially affect your ability to complete an assessment task to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, go to <http://www.latrobe.edu.au/special-consideration>

STUDENTS WITH A LEARNING ACCESS PLAN (LAP)

Students who have a Learning Access Plan must apply for any adjustments they may need via the same channels but may have access to more flexible arrangements depending on their circumstances. Please refer to the [Assessment Procedure – Adjustments](#) and your LAP.

REVIEW AND RE-MARK

For information on what to do if you wish to seek a review of an assessment outcome [see Part B](#) of the Validation and Moderation Procedure.

ACADEMIC INTEGRITY

Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating.

The penalty for submitting an assignment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to the **Academic Integrity - Schedule of Penalties and Actions** within the [Student Academic Misconduct Policy](#).

You should understand and be able to practice integrity in all your academic work, be familiar with the [Academic Integrity website](#) and complete the **Academic Integrity Module (AIM)** found in your LMS.

If you have any specific questions about **academic integrity** and your assessment, your Subject or Course Coordinator will be able to assist.

STUDENT FEEDBACK ON SUBJECT SURVEYS

The **Student Feedback on Subjects (SFS) Survey** is one of the quality assurance processes that occur across the University. In this survey, you can tell us about your learning experiences in this subject. We value your feedback. Your views are taken seriously and assist us to improve the learning experience for future students through the LTU Subject Monitoring process.

The surveys will be available for you to complete towards the end of the subject teaching period. Feedback from previous students who completed this subject is available in the LMS site for this subject.

For more information you may contact surveys@latrobe.edu.au.

CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe coursework student, you have rights and responsibilities which are governed by La Trobe statutes and regulations, as well as policies and procedures. La Trobe University aspires to provide an outstanding student experience which is conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students.

<https://policies.latrobe.edu.au/document/view.php?id=225>.

STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their university experience. A student complaint may be about an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the **Complaints and Feedback** page at: <https://www.latrobe.edu.au/students/support/complaints>.

GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing.

For the full range of services and facilities available for you, go to **Student Help and Support** at <http://www.latrobe.edu.au/students/support>.

STUDYING ONLINE

When you are in studying an online subject, please be sure that you can continue to access our learning support services through the Learning Hub (academic skills, English language, Maths and Science) via Zoom appointments that you can book by accessing the [Learning Hub LMS page](#) or by emailing the LearningHub@latrobe.edu.au.

You also have access to assignment feedback and online study support chat through Studiosity that can be accessed from any subject [LMS](#) page. You can also access the Library services remotely through [the library website](#).

LIBRARY LEARNING SERVICES

Your University Library offers more than a place to access information or a quiet place to study. We have a team of friendly specialists and a wide range of services available to assist you in your academic studies. Discover how by visiting your [Library Website](#) and the [Study and Research Support](#) webpage, and exploring the following options:

- **ACHIEVE@UNI**
<https://www.latrobe.edu.au/students/study-resources/learning/achieve> - Browse a collection of online learning resources, including writing samples from real students, to help you understand and complete your assessments.
- **PATHFINDER – YOUR GUIDE TO LEARNING SUCCESS**
<https://lms.latrobe.edu.au/course/view.php?id=57964>
Find practical, online advice to help you develop effective study habits, improve your assessments, prepare for exams, and enhance your research skills. Also, learn how to connect with friendly specialists for assistance including Academic Skills Language Advisors, Librarians, Peer Learning Advisors, Maths, Science and Coding Hub Tutors.
- **ENGLISH FOR SUCCESS – ACADEMIC ENGLISH TRANSITION**
<https://lms.latrobe.edu.au/course/view.php?id=137328>
Find strategies to help you transition to study in Academic English, whether you are an international student or a domestic student in your first year of study.
- **STUDIOSITY**
https://studiosity.com/connect/users/sign_in
An online writing feedback and study skill chat service available when you need it. Access this service through your Subject LMS page, or the Library's **Study and Research Support** webpage.
- **ASSESSMENT PLANNER**
<http://www.latrobe.edu.au/library/assessment-thesis-support/plan-my-assessment> – Create a personal assessment schedule.
- **EXPERT SUBJECT HELP GUIDES**
<https://latrobe.libguides.com/> - Find expert discipline and specialist help for your field of study.
- **REFERENCING AT LA TROBE UNIVERSITY**
<http://latrobe.libguides.com/referencing/> - Understand why, when and how to reference your work.
- **ACADEMIC REFERENCING TOOL**
www.lib.latrobe.edu.au/referencing-tool - Get detailed examples of referencing styles.
- **CHAT TO LIBRARY STAFF**
<https://www.latrobe.edu.au/library/about/contact> - Chat online with a specialist – not a chatbot – for advice on finding information, referencing, and more.
- **CONTACT US**
www.latrobe.edu.au/library/about/contact - Call or email us to learn how we can assist you.

OTHER LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. For more information, visit the **Learning and Support Programs** page at <http://www.latrobe.edu.au/students/study-resources/learning>.

ASK LA TROBE

ASK La Trobe is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe. ASK La Trobe can provide you with help relating to enrolments and timetabling, fees and financial support, exams, results and graduations & student life and support services.

Go to **ASK La Trobe – Current Students** at <http://latrobe-current.custhelp.com/app/home> and start by looking at our **FAQs** or visit us at your campus **ASK La Trobe Help Zone**.

If you can't find what you're looking for:

- **Call** 1300 La Trobe (1300 528 762)
- **Ask a question** online at <http://latrobe-current.custhelp.com/app/ask>
- **Visit the ASK La Trobe Help Zone.** Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing system**.

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HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful and many factors can affect your mental and physical health, in and outside of the University. Check out the services available to you at La Trobe to support your health and wellbeing at www.latrobe.edu.au/wellbeing.